



Usefulness of a microalgal biorefinery in conversion to chemical products and recent technology in automatic evaluation of microalgae

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A biorefinery is seen as a viable alternative to the petrochemical refinery. Microalgal biorefineries have a competitive advantage in sustainability because of their operation, which demands mainly light and inorganic substances. In addition to the choice of algal species, the maintenance and management of algal quality in the culture are critical challenges in a microalgal biorefinery. Therefore, automatic monitoring systems contribute immensely to the progress in the biorefinery. By comparing the features of an automated cell counting method with those of standard methods, this review highlights the features of the former method with fluorescence detection of chlorophyll. The method has the potential to guarantee the algal biomass in terms of quality and quantity, resulting in significant contribution to the biorefinery.

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Introduction

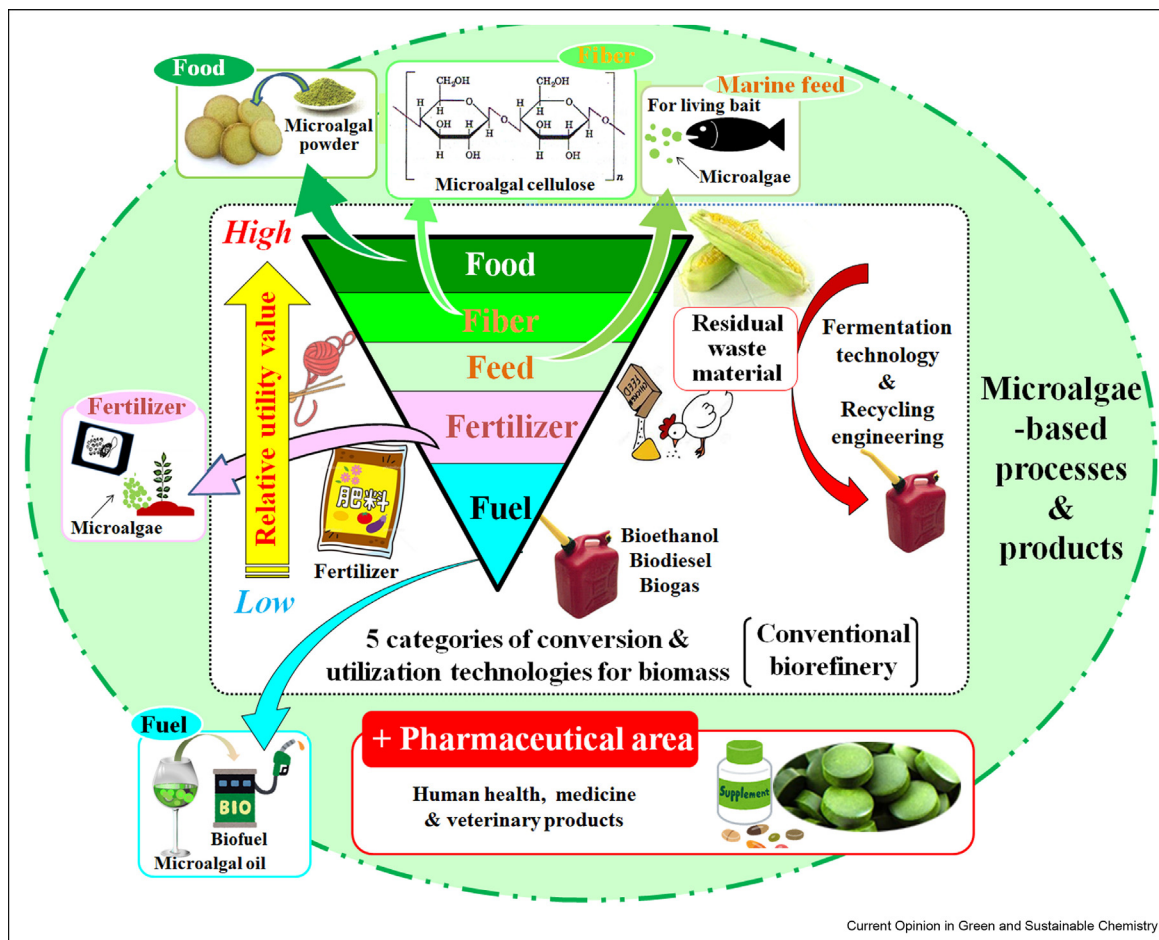
Considering the Sustainable Development Goals (SDGs) from the perspective of conservation of earth resources, an environmentally attractive biorefinery holds a key position in the green chemical refinery. The biorefinery especially contributes to ensuring the SDG nos. 9 and 12, which are for industry, innovation, and infrastructure and sustainable consumption and production patterns, respectively. There are many other kinds of bioeconomic technologies associated with the other SDG nos. 14 and 15, which are for life below

water and for life on land, respectively. To mitigate the global warming caused by the consumption of fossil resources, several methods of the biorefinery have been documented to date in bioeconomic conversion [1,2] and bioengineering conversion to valuable products [3–7].

Biomass-based feedstocks, which are not used for food or feed, are abundant because agricultural residues, forestry residues, energy crops, and other organic residues are constantly formed as the waste treatment stream. Although microalgal biomass has become an important feedstock for the production of third-generation biofuels [8–10], the algal residue—extracted lipid has been also produced as waste residue [11]. At this time, such a nonrecyclable organic portion of biosolids, sludge, and waste food is an attractive source of materials for the biorefinery. These feedstocks have been converted into intermediates in the production processes of sugars, chemical building blocks, bio-oils, and gaseous mixtures through deconstruction and fractionation processes. Then, these intermediates have resulted in primary end products used as renewable and sustainable sources. The development of the bioeconomy, however, requires building multi-integrated biorefineries, which are capable of converting a broad range of biomass feedstocks into affordable bioproducts, biopower, and biofuels [11,12].

Drawing attention toward the biomass-based biorefinery, five major categories of conversion and utilization technologies for biomass have been established in food, fiber, feed, fertilizer, and fuel areas (Figure 1). When utility value is shifted toward food from fuel, the biomass-based product has a relative utility value and becomes more diversified in commercial usage. Thus, a biomass-based product with a higher relative utility value can produce subordinate products in the residual wastes using both fermentation technologies and recycling engineering to fuel products. Instead of conventional biomass-based feedstocks (the central white region in Figure 1), microalgal biomass (the surrounding green area) is also expected of the foregoing five values in food [13], fiber [14], biofertilizer [15,16], biodiesel [8,17], and other energy source [17,18]. Microalgae can enhance the nutritional value of food and animal feed and play a crucial role in aquaculture and cosmetics [20].

Figure 1



Main categories of both conversion and utilization technologies for biomass. The white area is shown from conventional agricultural residues, forestry residues, energy crops, and other organic residues, while the green area is from microalgae.

Microalgal biorefinery

In addition to these five values, pharmaceuticals and health care are also important targets for the microalgal biorefinery [6,19,21,22]. The microalgal biorefinery, depending on light and inorganic materials, has a competitive advantage in sustainability.

Spherical eukaryotic microalgae species such as *Dunaliella salina*, *Chlorella* sp., *Haematococcus pluvialis*, and *Coelastrella striolata* var. *multistriata* have been used as valuable microalgae for industrial uses [23,24]. In addition to spherical microalgae, amorphous eukaryotic microalgae such as *Botryococcus braunii* [9] and others such as cyanobacteria and filamentous *Spirulina platensis* are also able to produce biofuels [9], as well as astaxanthin [24] and lutein [24].

To widen microalgal applicability for industrial application, there is a need to consider not only the selection of algal species but also the assessment and

maintenance of their quality. Compared with plant biomass, the assessment of microalgae using the naked eye is not easy in practice. Thus, easily missed characteristics, such as sensitivity of microalgae to environmental changes [22,25], ultimately determine the quality of each biomass-based product. Based on the practical situation, the routine control, maintenance, and management of algal quality in culture are critical challenges. A chlorophyll assessment based on either absorption spectroscopy or fluorometry allows a nondestructive measurement and can contribute highly to the routine assessment and management of microalgal growing.

Evaluation of microalgae

Conventional methods for the evaluation of microalgae include optical detection techniques such as microscopy-based hemocytometry [26,27], turbidity [28], and gravimetry. Table 1 summarizes both the

Table 1

Advantages and disadvantages of several standard methods for microalgae evaluation.

Method					
End point	Turbidity	Hemocytometry	Chlorophyll meter based on fluorometry	Flow cytometry	Laser scanning cytometry
Operability	Easy	Need of proficiency in operation	Easy	Need of proficiency in operation	Need of proficiency in operation
Nondestructive measurement	Nondestructive^a	Microassay	Nondestructive	0.2 ml ^a — a few milliliters of volumes	Microassay
Selectivity for algae	Low	Need of proficiency in operation	High	High^b	High^b
Measurement time	A short time	Depend on cell density (but longer than turbidity)	A short time	A short time of measurement (but need for several operations before use such as cleanup to the flow path)	Need for proficiency in operation (but not short time)
Visualization	Not applicable	Applicable as necessary	Not applicable	Not applicable	Applicable
High-resolution image	Not applicable	Not applicable in normal cases	Not applicable	Not applicable	Applicable
Multiparameter analysis	Not applicable	Not applicable	Not applicable	Applicable	Applicable
Capital investment spending	Reasonable cost	Low cost	Reasonable cost	Expensive	Quite expensive

The italics and bold letters mean an advantage on each end point.

^a Some devices are applicable to microassay.

^b To achieve a precise selectivity for microalgae, a sophisticated setup with respect to the target cell size and fluorescence properties of each device is needed.

advantages and disadvantages of each method. A chlorophyll meter, using fluorometry, was tuned for detecting chlorophyll [29,30] and could provide information about the abundance of phytoplankton. In an analogy with a chlorophyll meter, flow cytometry was adequate for chlorophyll autofluorescence and cell-by-cell analysis [26,31]. The dry weight method (gravimetry) was a simple but conventional and frequent method for evaluating the biomass of microalgae [32–34].

In addition to Table 1, Figure 2 summarizes the end points for each research method. Manual hemocytometry has the advantage of availability at low cost but the disadvantage of time wastage. High-precision determination requires operators with proficiency in hemocytometry to avoid user biases [35]. Moreover, it is difficult to diagnose cell status on appearance alone.

Turbidity can be used at low cost because a spectrophotometer for turbidity is the infrastructure equipment in a standard laboratory. In comparison with other methods, however, turbidity has disadvantages in terms of selectivity [36]. The concentration of the cells obtained by turbidity must be confirmed using a standard curve obtained by hemocytometry or other methods.

The value obtained by turbidity is generally specific to each species of the microorganism. Moreover, turbidity might not be able to determine whether the algal health is out of condition.

Requiring relatively large sample amounts (within the milligram scale), both the unsuitability for high-throughput research and the destructive analytical procedure might be disadvantageous for gravimetry. In addition to sample amounts, this method requires careful washing of the test sample to eliminate inorganic materials derived from the culture medium for an accurate measurement.

The contents of photosynthetic pigments in an algal cell vary widely in response to algal acclimatization to light conditions at the measurement points in the laboratory culture, outdoor photobioreactors, and nature [37–39]. To distinguish between target cells and noise signals, accurate evaluation of target cells by using flow cytometry requires sophisticated skills to set up measurement parameters. The dependence of the evaluation of pigment contents exclusively on fluorometry and flow cytometry might lead to the misinterpretation of the character of microalgae in culture for lack of visual information [39].

Figure 2

Method Endpoint	Suitability for high-throughput analysis					
	An automated cell counter with fluorescence detection	Chlorophyll meter	Turbidity measurement	Flow cytometry	Hemocytometry	Dry weight measurement
Measurement accuracy	⊙	⊙	○	○	○	○
Handleability	⊙	⊙	⊙	○	⊙	×
Sensitivity	⊙	⊙	△	⊙	○	×
Measurement cost	○ or ⊙*	⊙	⊙	○ ⁺	⊙	○
Microassay	⊙	×	○ ⁺	○ ⁺	⊙	×
Diagnosis of an algal status	⊙	△**	×	⊙	△	×
Automated visualization	⊙	×	×	×	×	×
Automated graphical output	⊙	×	×	⊙	×	×
Compact space-saving design	⊙	⊙	△ ⁺	△ or ○ ⁺	⊙	⊙
Capital-investment spending	○	○	○	×	⊙	⊙

Possible ← Potential of (semi-) automatization of a production line for algal biorefinery → Difficult

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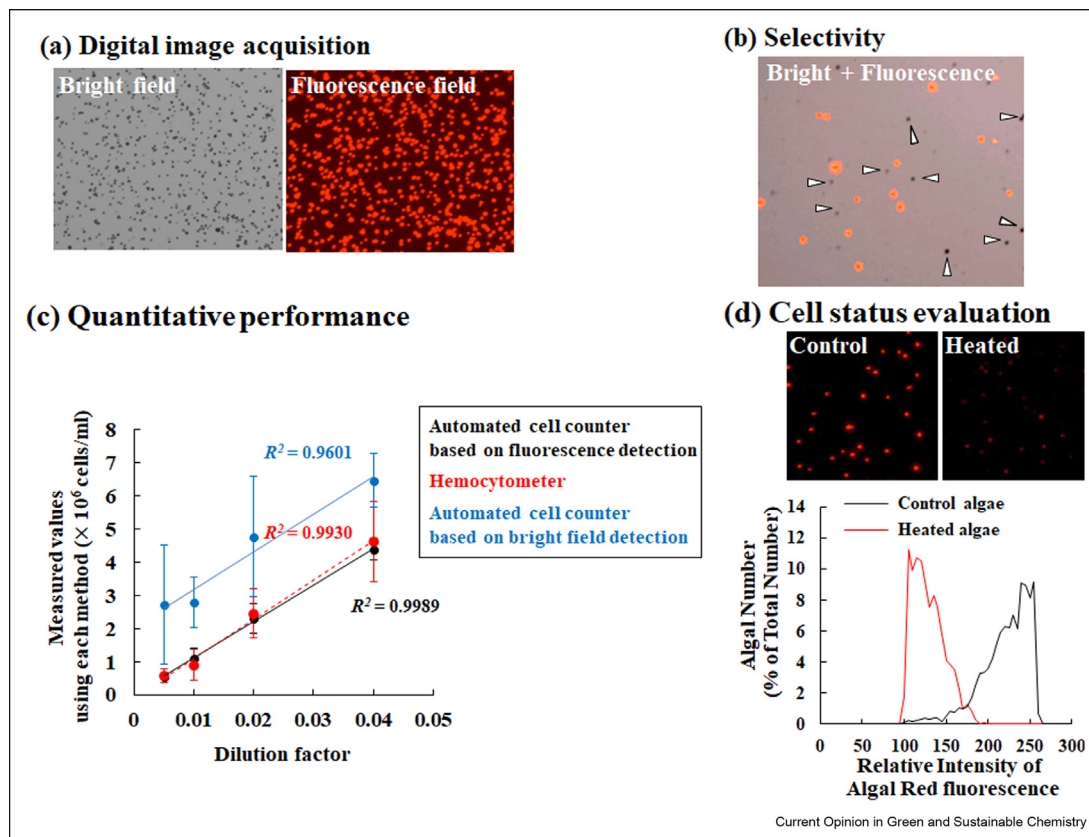
Comparison of the functions and features of an automated cell counter device with several standard methods to evaluate microalgae. Adequateness of high-throughput analysis and automatization of algal biorefinery is shown respectively in the upper side and the lower side of the table. +: this is contingent on a device. *: the cell counter device can use a disposable as well as reusable slide. **: measurement using a chlorophyll meter requires relatively larger volume than that required by the microscopic observation. The device evaluates algal population in culture rather than a single cell.

A new approach to evaluate microalgae for *in situ* and semiautomatic management

Measuring the accuracy, sensitivity, and availability with appraisal cost becomes an important criterion in culture management in addition to nondestructive analysis or microanalysis. In the studies by Takahashi [22,39], an automated cell counter, Countess™ II FL automated cell counter (Thermo Fisher Scientific Inc., Waltham, MA, USA), enabled the detection of chlorophyll fluorescence using a filter Cy@5 of EVOS light cubes [40]. Figure 2 summarizes the features of the cell counter device. In an analogy with a chlorophyll meter and flow cytometer, chlorophyll fluorescence was noninvasive, selective, and sensitive in the measurement without any specific reagent, resulting in no wastage of algal biomass. A chlorophyll-based method was broadly applicable to a wide variety of microalgae. In addition to the detection

of chlorophyll fluorescence, the cell counter device can determine the cell density automatically, based on hemocytometry. These features might make hemocytometry preferable as it helps to avoid time wastage and user bias through the automated algorithmic analysis. In addition, this can help to avoid the problems associated with both fluorometry and flow cytometry as it does not rely on visual information. Laser scanning cytometry also provides the fluorescence properties of an individual cell along with high-resolution visual information for clinicians and scientists [41]. A laser scanning cytometer is generally more expensive than standard analytical instruments. Although an automated cell counter with fluorescence filters falls short of laser scanning cytometry in imaging resolution, the cell counter has two functions similar to both hemocytometry and flow cytometry.

Figure 3



Example of measurement using an automated cell counter with the fluorescent filter for chlorophyll detection. Here, all panels were referred from the study by Takahashi [22] and modified.

The automated cell counter with the fluorescence filter for microalgal evaluation

To conclude, as seen in Figure 3, the cell counter device can visualize the target algae, determine their quantity, and concurrently evaluate their cell status based on chlorophyll integrity. Upon excitation, the chlorophyll in microalgae emits red fluorescence at approximately 680 nm [39,42]. Using the filter for Cy5, the cell counter device could detect all tested microalgae in the bright and corresponding fluorescence fields (Figure 3a) [22,39].

Figure 3b is an image of the measurement of the green alga, *Parachlorella kessleri*, obtained by using the cell counter device. The red image shows microalgae, whereas arrowheads are bubbles and other debris other than algae. As expected, the utilization of chlorophyll fluorescence resulted in the selection of target microalgae with the determination of false recognition in the fluorescence image. Therefore, without fluorescent selection, several particles such as bubbles and other debris were detected along with

microalgae (the blue line in Figure 3c). In contrast, the automated measurement of this device with fluorescent selection for chlorophyll (black line) was statistically of the same level of accuracy as that of manual hemocytometry (the red line in Figure 3c) [22,39].

Chlorophyll integrity can also deteriorate owing to changes in temperature, pH, heat, and other uncertain factors [25]. In the study by Takahashi [22,39], heat reduced the fluorescence of microalgal chlorophyll (the inset images in Figure 3d) because of the vulnerability of the chromophore to heat. When the pigment contents were evaluated for cell health, the indicator was superior in both cost performance and availability for use in routine work. Figure 3d shows chlorophyll integrity with (red line) or without heat treatment (black line). Thus, the device recognized the rapid changes in each algal status due to changes in chlorophyll integrity.

Figure 2 compares the differences among the methods. Beyond a chlorophyll meter, turbidity, flow cytometry, hemocytometry, and dry weight measurement, the cell

counter device simultaneously guarantees an algal biomass in terms of both quality and quantity. In particular, the characteristic features of the device described previously indicate the possibility of a complete automation of the algal biomass management as a major benefit. As a result, the automated cell counter might recognize many types of industrial microalgae having a unicellular spherical shape by using the present algorithm of the automated cell counter.

Conclusion

This review highlighted that microalgal biomass and their biorefinery showed potential for contributing to several industries including food, fiber, and chemicals. Being too small for material feedstocks, an increase in microalgal numbers (microalgal yields) per unit volume takes precedence over everything else. In addition to cell yields, algal quality is directly linked to the yield, quality, and stability of the end product from algal biomass. Therefore, the quality assurance of an algal biomass is a great challenge in terms of both quality and quantity.

Comparing several features of the standard methods for the evaluation of microalgae, as seen in Figure 3, the features of the automated cell counter device with fluorescence detection of chlorophyll ensured the reliability of measurement accuracy in cell counting equivalent to hemocytometry. The cell counter device has some competitive advantages in terms of sensitivity to detect microalgae, measurement of cost, evaluation of algal health, capital investment, and so on (Figure 2). The aforementioned points imply that the cell counter device might make a significant contribution to algal management in culture and the algal biorefinery.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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